Software Requirements Specification

for

Designing a Privacy-Preserving, Offline Machine Learning System for Phishing Detection Using URL Features

Version 2.6 approved

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Executive Summary

This Software Requirements Specification (SRS) document defines the design and functional scope of a machine learning-based phishing detection system that classifies URLs as either malicious or legitimate using URL-string-derived features exclusively. The system is designed as a standalone, offline pipeline, with no reliance on external services, live network queries, or third-party reputation systems during execution.

The system supports the study of phishing detection using URL-derived features under strict operational constraints, including air-gapped deployment, privacy preservation, and independence from external data sources. While existing phishing detection systems often achieve strong performance, many depend on supplementary inputs such as Domain Name System (DNS) queries, WHOIS data, webpage content analysis, or cloud-based threat intelligence feeds. These dependencies introduce limitations related to data exposure, reproducibility, and deployability in restricted environments.

To investigate this problem, the system extracts a defined set of lexical features from URLs and applies multiple supervised machine learning classifiers. The classification outputs are aggregated using a majority voting ensemble mechanism, enabling comparative analysis of individual model performance and overall system resilience. The pipeline supports end-to-end experimentation, including dataset construction, preprocessing, feature extraction, model training, and cross-dataset evaluation.

The primary contribution of this work is the development of a constrained experimental framework for the systematic evaluation of phishing detection performance in a fully offline, privacy-preserving, and lightweight environment. Rather than introducing a novel algorithm, this project focuses on analyzing the effectiveness and limitations of URL-only detection when external data sources are intentionally excluded.

This SRS document outlines the system's functional requirements, data structures, processing workflow, and operational constraints. The implementation is intended as a research-oriented prototype designed to support empirical evaluation, reproducibility, and controlled experimentation. The results of this system aim to contribute to cybersecurity research by examining the trade-offs between detection accuracy, generalization, privacy preservation, and system independence.

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